PROPER BASE CHANGE FOR SHEAVES OF SETS ON TOPOLOGICAL SPACES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof ends with the author's own 'Some details omitted.' after the reduction; the sentence is reproduced verbatim. The two reductions are named explicitly and the only input used is one whose first part is checked to work for sheaves of sets, so the omission is bounded.

Editorial note. The counted spans carry 4 navigation stubs and no bibliography citation.

Editorial note. The preamble is the chapter's retained minimal preamble (sources/stacks-cohomology/retained-preamble.tex lines 1-81), a verbatim excerpt of the native preamble.

Editorial note. The label lemma-proper-base-change-sheaves-of-sets is defined only in cohomology.tex, so no cross-chapter restatement had to be weighed for this card.

Allowed standard prerequisites (editorial).

- Proper base change for complexes of abelian sheaves. Same chapter, Theorem theorem-proper-base-change and Lemma lemma-proper-base-change, whose proofs are followed at proof lines 3157-3161. Supplies the reduction to the stalk formula and the shape of the argument.
- The zeroth degree case of the cohomology of a closed subset, valid for sheaves of sets. Same chapter, Lemma lemma-cohomology-of-closed, invoked at proof lines 3161-3163. Supplies the identification of the sections over the fibre.

Source reference. Original lemma 09V3: sources/stacks-cohomology/pinned-original-source.tex, source lines 2848–2848. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 09V5: sources/stacks-cohomology/pinned-original-source.tex, source lines 3059–3059. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original theorem 09V6: sources/stacks-cohomology/pinned-original-source.tex, source lines 3101–3101. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Date: stacks-project:cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Lemma 0.1 (Proper base change for sheaves of sets)**.** *Consider a cartesian square of topological spaces*

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ Y' & \xrightarrow{g} & Y \end{array}$$

Assume that f is proper. Then $g^{-1}f_*\mathcal{F} = f'_*(g')^{-1}\mathcal{F}$ for any sheaf of sets $\mathcal{F}$ on X.

Proof. We argue exactly as in the proof of Theorem 09V6 and we find it suffices to show $(f_*\mathcal{F})_y = \Gamma(X_y, \mathcal{F}|_{X_y})$. Then we argue as in Lemma 09V5 to reduce this to the $p = 0$ case of Lemma 09V3 for sheaves of sets. The first part of the proof of Lemma 09V3 works for sheaves of sets and this finishes the proof. Some details omitted. □